CHRIS JOSEPH GEORGE

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About Me

Computer Science graduate specializing in **Artificial Intelligence and Machine Learning**, with hands-on experience building AI applications, deep learning models, and backend APIs. Proficient in **Python, machine learning, FastAPI, and model development**, with project experience in **Generative AI, predictive maintenance, medical image classification, and explainable AI**. Co-author of an **IEEE-published research project** involving an AI-powered aircraft engine digital twin. Interested in developing reliable AI solutions and deploying practical, data-driven applications.

Skills

- **Programming Languages**: Python, SQL, Java
- **AI, ML & Deep Learning**: Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, SHAP, LIME, Explainable AI
- **Generative AI & Agentic AI**: RAG, Agentic AI, LangGraph, LLM Integration, Sentence Transformers
- **Backend Development & Deployment**: FastAPI, Flask, Streamlit, REST APIs, Google Cloud Platform (GCP)
- **Databases & Tools**: PostgreSQL, MySQL, SQLite, MongoDB, Git, GitHub

Projects

- **FastAPI Technical Documentation Assistant – Self-Corrective RAG**
  Python, FastAPI, LangGraph, Gemini, ChromaDB, Sentence Transformers, Streamlit
  - **Agentic RAG Pipeline**: Engineered a self-corrective Retrieval-Augmented Generation (RAG) pipeline using LangGraph, combining document retrieval, relevance grading, query rewriting, and response regeneration.
  - **Retrieval & Grounding**: Implemented ChromaDB vector search with local Sentence Transformer embeddings and LLM-based relevance evaluation to improve retrieval quality and ground generated answers in retrieved context.
  - **Response Validation**: Integrated groundedness checks to evaluate generated responses against retrieved documents and return source metadata with answers.
  - **API & Integration**: Developed FastAPI endpoints for querying, document ingestion, document management, and user feedback, with support for Markdown, text, HTML, and URL inputs. Integrated Tavily web search as a fallback and built a Streamlit interface with session-based conversation memory.
  - **GitHub**: github.com/CJG1410/fastapi-rag-assistant
- **AI-Powered Digital Twin for Aircraft Engines – IEEE Published**
  Python, Machine Learning, SHAP, Federated Learning, Streamlit, Unity, Raspberry Pi
  - **Predictive Maintenance**: Developed an aircraft-engine digital twin framework using machine learning to estimate Remaining Useful Life (RUL) and monitor engine health.
  - **Explainable Predictions**: Integrated SHAP to interpret model predictions and identify features influencing estimated engine health and maintenance needs.
  - **Federated Learning**: Implemented a Federated Averaging (FedAvg) simulation to explore distributed model training without centralizing all training data.
  - **Digital Twin Interface**: Connected a Streamlit monitoring dashboard with a Unity-based visualization environment for presenting engine health, predictions, and simulation results.
  - **Research Publication**: Published the project in IEEE conference proceedings: IEEE Xplore (Document 11375942)
- **MediScanNet – Multi-Stage Chest X-ray Classification**
  Python, PyTorch, DenseNet121, ResNet50, CUDA, Streamlit
  - **Deep Learning Architecture**: Designed a dual-stream image classification model combining DenseNet121 and ResNet50 to learn complementary features from chest X-ray images.
  - **Multi-Dataset Training**: Developed a three-stage training pipeline using NIH ChestX-ray14, RSNA Pneumonia Detection, and CheXpert datasets.
  - **Ensemble Classification**: Integrated specialist networks with soft-voting aggregation to combine model predictions for chest X-ray classification.
  - **Training Optimization & Interface**: Applied PyTorch Automatic Mixed Precision (AMP) and GPU Tensor Cores to optimize training, and developed a Streamlit interface for displaying predictions and confidence scores.
- **AI-Based Context-Aware Route Optimization System**
  Python, Scikit-learn, Random Forest, Streamlit, Google Maps API, Folium
  - **Context-Aware Route Evaluation**: Developed a route recommendation system that considers traffic conditions, air quality index (AQI), and fuel-efficiency metrics when comparing travel routes.
  - **Predictive Route Scoring**: Built a Random Forest regression model to estimate route scores and support context-aware route comparisons.
  - **Interactive Visualization**: Integrated map-based route visualization and analytics using Folium and Streamlit.
  - **GitHub**: github.com/CJG1410/AI-Based-Context-Aware-Route-Optimization-System

Experience

- **Chillar Payment Solutions Pvt Ltd** — Kerala, India
  Software Development Intern — *May 2024 – June 2024*
  - **Database Optimization**: Optimized SQL queries through indexing and query refactoring, improving query performance by 35%.
  - **Backend API Development**: Developed REST APIs using FastAPI to support backend integration, reducing system integration time by 25%.
  - **Testing & Reliability**: Created automated unit tests, increasing code coverage by 40% and improving backend code reliability.
- **Cisco AICTE Virtual Internship Program** — Remote
  Networking Intern — *May 2024 – June 2024*
  - **Network Simulation**: Designed and simulated network topologies using Cisco Packet Tracer to practice network configuration and troubleshooting.
  - **Networking Fundamentals**: Completed training in networking and network security, covering routing, switching, and network protocols.

Volunteer Experience

- **Google Developer Student Clubs (GDSC Karunya)** — Coimbatore, India
  Google Cloud and Cybersecurity Facilitator — *Nov 2023 – Feb 2024*
  - **Technical Workshops**: Delivered 5+ workshops on Google Cloud and cybersecurity, introducing 100+ students to core concepts and practical exercises.
  - **Student Mentorship**: Mentored 20+ students in technical learning and certification preparation.
- **E-Cell Karunya – Entrepreneurship Cell** — Coimbatore, India
  Senior Corporate Relations Coordinator — *Nov 2022 – Jan 2025*
  - **Corporate Relations**: Built corporate partnerships and supported sponsorship outreach, contributing to INR 5,00,000+ in sponsorships for entrepreneurship initiatives.
  - **Event Outreach**: Supported marketing and outreach initiatives to promote entrepreneurship events and increase student participation.
- **KarunyaHacks – University Hackathon Initiative** — Coimbatore, India
  Core Team Member — *Oct 2022 – Apr 2024*
  - **Hackathon Coordination**: Supported the planning and execution of three university-level hackathons involving 200+ participants.
  - **Technical Workshops**: Helped organize eight workshops on AI and web development, reaching 150+ students.
  - **Event Operations**: Coordinated event logistics and resource allocation to support smooth hackathon execution.

Education

- **Karunya Institute of Technology and Sciences** — Coimbatore, India
  Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) — *Aug 2022 – Jul 2026*
  - **Relevant Coursework:**: Database Management Systems, Artificial Intelligence, Machine Learning Techniques, Software Engineering, Generative AI, Machine Learning with TensorFlow, Robotic Process Automation
- **Sacred Heart CMI Public School** — Ernakulam, India
  High School — *Jun 2014 – May 2022*

Certifications

- Google Cybersecurity Professional Certificate [Link]
- Networking Essentials - Cisco Networking Academy [Link]
- Introduction to Artificial Intelligence - IBM [Link]
- Java Programming Fundamentals - Infosys Springboard [Link]
- PCAP Programming Essentials in Python - Cisco Networking Academy [Link]

Languages

- **English**: Fluent
- **Malayalam**: Native Speaker